

The Impact of Cognitive Biases on the Believability of Fake News

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Abstract

Modern technologies, especially social networks, contribute to the rapid evolution and spread of fake news. Although the creation of fake news is a serious issue, it is the believability of fake news and subsequent actions that produce negative outcomes that can be harmful to individuals and society. Prior research has focused primarily on the role of confirmation bias in explaining the believability of fake news, but other biases are likely. In this research, we use theories of truth and a taxonomy of 10 cognitive biases to conduct an exploratory, qualitative survey of social media users. Five cognitive biases (herd, framing, overconfidence, confirmation, and anchoring) emerge as the most influential. We then propose a *Cognitive Bias Mitigation Model* of methods that could reduce the believability of fake news. The mitigation methods are grouped according to three themes as they relate to the five biases.

Keywords: fake news, cognitive bias, social media, herding bias, framing bias, overconfidence bias, confirmation bias, anchoring bias, Cognitive Bias Mitigation Model

'The confidence people have in their beliefs is not a measure of the quality of evidence but of the coherence of the story the mind has managed to construct.' Daniel Kahneman

1. Introduction

The term, *fake news*, gained prominence during the 2016 U.S. presidential election (Khan et al., 2022, Kim et al., 2019, Kim and Dennis, 2019) and has continued to displayed disruptive potential in managing public perceptions during a crisis such as the COVID-19 pandemic (Chakravorti, 2020, Shirish et al., 2021). Fake news is one of several terms (e.g., misinformation, disinformation, rumors) that have been used interchangeably (e.g., Wu et al., 2022) to describe false information (Kapantai et al., 2021). Fake news is a specific type of disinformation that contains verifiably false news stories (Allcott and Gentzkow, 2017, Lazer et al., 2018). It consists of false or manipulated information presented as a news story with the intention to deceive the consumer. Fake news has received significant attention by researchers who have studied its creation, consumption, dissemination, and detection (Beauvais, 2022). While the existence of fake news is problematic, fake news is most concerning when it is believed and acted upon in a way that leads to undesirable outcomes.

The consumption and spread of fake news have been primarily attributed to confirmation bias and believability (Moravec et al., 2020). Confirmation bias is a cognitive bias that causes a consumer of new information to believe it is true if it aligns with their current beliefs or ideology (Kim et al., 2019).¹ While confirmation bias is the most prevalent of the cognitive biases studied regarding believability of fake news (George et al., 2021), there are likely additional cognitive biases that could influence the believability of fake news and the formation of new ideological beliefs. This research addresses this gap by conducting an exploratory study to identify additional cognitive biases that may influence believability.

The questions addressed in this research are: *What biases affect the believability of fake news?* and *What mitigation methods may be able to counter these biases to reduce the believability of fake news?* The objectives are to: 1) identify the cognitive biases that influence the believability of fake news; and 2) identify cognitive bias mitigation methods that can be applied within a fake news context to minimize the

¹ This implies that the consumer of information has prior knowledge of, or a belief system about, the topics being consumed.

impact of the identified cognitive biases on the believability, and subsequent spread, of fake news. We explore and identify cognitive biases through a qualitative survey that is analyzed using a taxonomy of ten biases. From the survey results, five cognitive biases emerge that can be used to guide strategies for fake news mitigation on social media. The contributions of this research are to understand the role of cognitive biases in fake news and propose mitigation strategies based on these biases.

Section 2 reviews relevant literature on fake news, believability and truth, and cognitive biases. Section 3 describes the research design for this exploratory study utilizing a survey with open-ended questions. Section 4 analyzes and interprets the survey results to identify cognitive biases that influence the believability of fake news and truth-seeking behaviors by social media users. Section 5 proposes a model for mitigating cognitive biases with recommendations to guide future research and Section 6 concludes the paper.

2. Related Research

This research investigates the believability of fake news. Believability is the noun form of the word believable, defined as '*capable of being believed especially as within the range of known possibility or probability*' (Merriam-Webster, 2022). Synonyms include credible, presumptive, plausible, and thinkable. At the root of these terms are beliefs and perceptions of truth (perceived truth), which are intertwined (Shah, 2003). As individuals consume fake news, they internalize its content and assess its believability, which can be influenced by cognitive biases as illustrated in Figure 1.

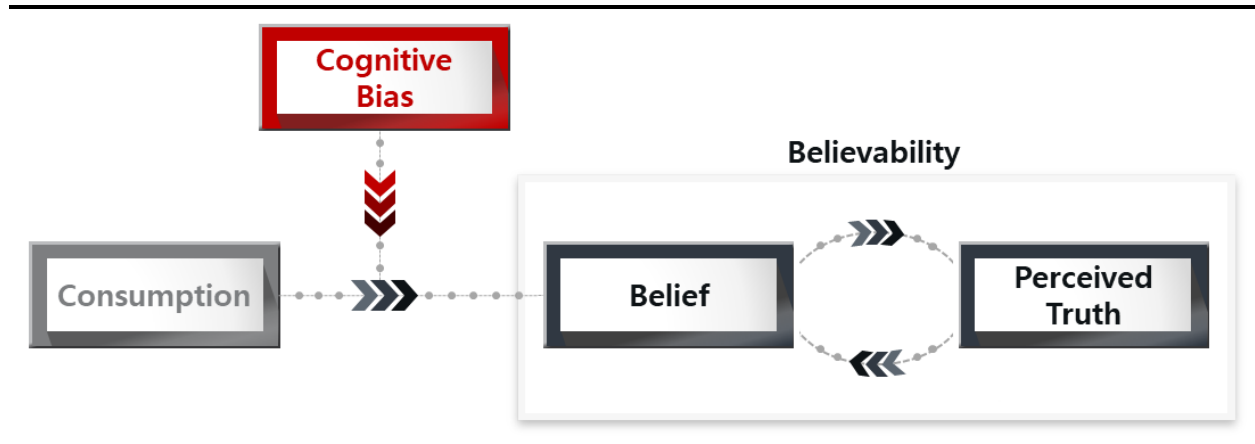


Figure 1. Relationship between consumption and believability

2.1 Fake News

As social media replaces or supplements traditional media outlets (Riemer and Peter, 2021, Ren et al., 2022), issues of credibility and consistency become complicated (Zhang and Li, 2020). These effects are further exacerbated due to network effects as false information has been shown to spread faster than legitimate news through the Internet and social media (Clark, 2020, Vosoughi et al., 2018). Various terms have been used to describe false information containing deceptive properties such as misinformation, disinformation, and fake news (Kapantai et al., 2021).

Two broad terms, misinformation and disinformation, are used to describe the occurrence of false information that is disseminated for others to consume. (Guo et al., 2020, Stahl, 2006). These terms differ only by intention. Misinformation is any false information (i.e., misleading, inaccurate, incomplete, etc.) that is unintentionally created and disseminated to others (George et al., 2021, Stahl, 2006). Alternatively, false information that is created deliberately with the intention to deceive or mislead others is referred to as disinformation (Guo et al., 2020, Stahl, 2006, Wu et al., 2022). While disinformation has deceitful and disruptive capabilities, the source of disinformation varies. Broadly, disinformation can come from anybody who intentionally disseminates false information, such as journalists, politicians, business leaders, or regular people who post on social media. Figure 2 depicts the categories of false information and how they are differentiated.

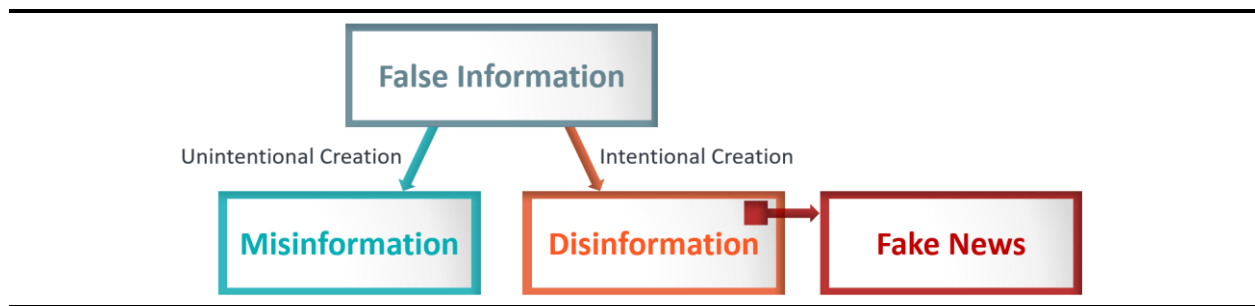


Figure 2. Categories of False Information

While the terms disinformation and fake news are sometimes used interchangeably (Wu et al., 2022), the term *fake news* is specific in relation to its format, who creates it, and how it is consumed. Fake news is verifiably false information that is intentionally created to appear as a news article designed to deceive the consumer (Allcott and Gentzkow, 2017). Fake news threatens the objectivity and creditability of real news and seeks to influence public opinion and adversely affect society (Zhao et al., 2020). Fake news articles, like other types of disinformation, contain deceptive properties that can mislead or manipulate individuals, thereby influencing opinions, perceptions, and actions (Moravec et al., 2020, Vogel et al., 2020). Although there is no fool proof system to prevent disinformation and its various forms, steps can be taken to mitigate its effects (Vasist and Krishnan, 2022). There is a wide variety of individuals who may create or share disinformation (i.e., friends, family, celebrities, journalists, etc.). However, we focus on third party news sources to evaluate whether consumers can differentiate real news from fake news. In this sense, we are not attempting to prevent all forms of disinformation, but rather identify methods to mitigate the effects of one specific type of disinformation; namely, fake news.

2.2 Believability and Truth

The concepts of truth and belief are intertwined, as evidenced when individuals consider whether they should believe a news story (Shah, 2003). Fake news is *verifiably false*, which is objective in nature. However, research studies often evaluate the *believability* of fake news (Moravec et al., 2020, Moravec et al., 2022), which can be subjective (based on beliefs and perceptions of truth) (Field, 2012). The result is a

disconnect between the identification of fake news (based on objective facts) and belief in fake news by a consumer (based on subjective views of truth).

Facts are independent of human cognition. Beliefs and perceptions of truth are often formed at a *personal* level based on experiences and interpretations; or at a *constructed* level based on shared meaning (Field, 2012). Personal levels of truth have been addressed through coherence and correspondence theory. Coherence theory states that individuals evaluate truth based on internal consistency and logical standards. Correspondence theory asserts that individuals seek evidence to support beliefs before making decisions about whether something is true (Mingers and Standing, 2020). The collective of personal truths among a group may result in a constructed truth, which can be explained by consensus theory, in which truth is established from agreement among members of a group or community (Mingers and Standing, 2020).

The boundaries of truth are often challenged between personal and constructed truth (Field, 2012). Limitations in our mnemonic capabilities can result in inaccurate memories or understanding of events (Autore and Ryan, 2020). Memory-based information processing is influenced by cognitive biases that transition objective observations into subjective interpretations (Hilbert, 2012). These inaccuracies may influence recollection and personal perceptions of truth when evaluating new information. Thus, when considering the subjective nature of perceptions of truth in coherence theory, cognitive biases influence believability. If personal truth perceptions are biased, the effects can extend to the group level, as suggested by consensus theory, resulting in herding effects. Prior research on fake news has recognized the effects of confirmation bias (Kim and Dennis, 2019, Kim et al., 2019, Moravec et al., 2019) and, to a lesser extent, herding bias (Celliers and Hattingh, 2020, David-Barrett, 2020). However, it is reasonable to assume there are other cognitive biases that could influence the believability of fake news.

2.3 Cognitive Bias

Cognitive biases are deviations from rational behavior (Haselton et al., 2015) that influence an individual's deception detection capabilities (Millar and Millar, 1997). From a multi-disciplinary review, George et al (2021) conclude that confirmation bias is the only cognitive bias currently considered as influencing the spread of fake news. With confirmation bias, individuals tend to favor information that confirms their pre-

existing opinions and ignore information that challenges these opinions (Kim et al., 2019). It is a significant factor that impedes deception detection, resulting in the consumption and continued spread of fake news; moreover, measures to mitigate confirmation bias by attempting to influence people's beliefs have limited results (Moravec et al., 2019).

The literature is replete with references to cognitive biases but lacking an appropriate list from which to investigate fake news. We, therefore, turned to the practitioner literature that identifies ten cognitive biases as influencing a person's reasoning in decision-making (CFI, 2020). The list comes from a financial context; a common domain for fake news discussions (Clarke et al., 2020, Costa et al., 2017). However, these biases have been observed in many other domains. The ten biases are described in Table 1 (along with additional references) and serve as a starting point for our exploratory investigation into the role that biases may play in the believability, and subsequent spread, of fake news.

Table 1. Types of Cognitive Biases

Term	Description
Anchoring Bias	Human tendency to overvalue a single piece of information (the anchor) and undervalue additional information when making judgments (Ahsen et al., 2019, p. 99)
Confirmation Bias	Human tendency to favor information that confirms one's pre-existing opinions — and ignore information that challenges them (Kim et al., 2019, p. 936)
Framing Cognitive Bias	Decisions made because of the way information is presented, rather than based on facts (CFI, 2020, Huerta et al., 2012, p.318)
Herd Mentality	Social dynamic in which individuals follow the actions of their peers (Jiang et al., 2018, p. 87)
Hindsight Bias	Tendency to overestimate one's prior knowledge of a fact or event after learning the actual fact (Pohl et al., 2018, p. 278)
Loss Aversion	Individuals weigh losses heavier than gains in making decisions; tendency to fear losses and avoid them (CFI, 2020, Lee and Joshi, 2017, p. 740)
Narrative Fallacy	False sense of comfort from limited data observations, leading to wrong conclusions (Nafday, 2009, p. 192)
Overconfidence Bias	Individuals' subjective belief of knowledge levels (i.e., the depth and breadth, relevance, and accuracy of knowledge) becomes incommensurately higher than their objective knowledge levels (Park et al., 2013, p. 1052)
Representativeness Heuristic	Judgement of similarity; when people falsely believe that if two objects are similar, they are also correlated with each other (CFI, 2020, Pitts and Browne, 2007, p. 94)
Self-serving bias	People are more likely to attribute positive events to themselves but dismiss negative events as attributable to other causes. (Mezulis et al., 2004, p. 711)

2.3.1 Anchoring Bias. Anchoring bias is the tendency to overvalue a single piece of information and can occur when irrelevant information is shown to someone prior to making a decision, resulting in a biased or bad decision (Downen et al., 2019).

2.3.2 Confirmation Bias. Confirmation bias occurs when people believe information that aligns with their pre-existing beliefs and ignore information that challenges them (Kim and Dennis, 2019). Studies evaluating the prevalence of confirmation bias in fake news found that users were more likely to believe and spread articles that aligned with their beliefs (Moravec et al., 2020, Moravec et al., 2022).

2.3.3 Framing Cognitive Bias. Framing cognitive bias occurs when people make decisions because of the manner in which information is presented, rather than based on facts (CFI, 2020, Huerta et al., 2012). When identical information is presented in different ways, the resulting decisions can be inconsistent (Tversky

and Kahneman, 1985). For fake news, this bias results from making judgements based solely on the coherence of a story without seeking corresponding, verifying information.

2.3.4 Herd Mentality. Herd mentality is the social dynamic that occurs when individuals follow the actions of their peers (Jiang et al., 2018). Herding occurs when individuals in a group align in their thoughts or behaviors (Raafat et al., 2009). It relates to consensus truth; individuals believe information presented by the herd, rather than seeking corresponding evidence.

2.3.5 Hindsight Bias. Hindsight bias occurs when individuals over-estimate their prior knowledge about facts or events once they know the actual outcome (Pohl and Erdfelder, 2017), suggesting that they “knew it all along” (Fischhoff, 1975).

2.3.6 Loss Aversion. Loss aversion relates to the idea that people tend to place more value on giving up or losing an object than on acquiring it (Kahneman and Tversky, 2013). Changes that make things worse (losses), seem more important than changes that make things better (gains) (Kahneman et al., 1991). Loss aversion may play a role in news and information searches (Schnellenbach, 2017).

2.3.7 Narrative Fallacy. Narrative fallacies occur when someone overestimates what happens in one situation and over-generalizes it to all situations (Taleb, 2007). Individuals perceive non-existent patterns in data and focus disproportionately on failures or successes (Nafday, 2009). Fallacies have been shown to affect the believability of fake news (Beisecker et al., 2022)

2.3.8 Overconfidence Bias. Overconfidence bias occurs when an individual’s subjective belief of knowledge levels (depth and breadth, relevance, and accuracy) becomes incommensurately higher than their objective knowledge levels (Park et al., 2013, p. 1052). It occurs when people place an inappropriate amount of confidence in their beliefs and judgements (Russo and Schoemaker, 1992).

2.3.9 Representativeness Heuristic. Kahneman et al. (1982) identified the representative heuristic to explain why individuals believe that a sub-sample of information is “representative” of the population, thereby distorting their estimate of true probabilities.

2.3.10 Self-serving Bias. Self-serving bias occurs when individuals alter their perception of causality to protect their self-esteem (Miller and Ross, 1975). Successes are attributed to one's own abilities; failures are blamed on external forces (Hastorf et al., 1970).

3. Methodology

We employed an exploratory qualitative methodology (Sarker et al., 2013) in which we used structured surveys to: 1) explore perceptions and sharing habits of fake news on social media; and 2) identify the cognitive biases that influence the believability of fake news during consumption. The recommended approach for conducting a qualitative survey involves eliciting responses from participants; reduction through data selection, coding, and clustering; and analysis to produce knowledge and understanding of a phenomenon (Romano Jr et al., 2003). We administered a survey to social networking users and evaluated their consumption and believability of fake news, based on their willingness to share the news with others. Cognitive biases that influence believability during consumption were identified by analyzing the respondents' sharing habits and perceptions of fake news. Given that individuals struggle to identify their own biases (Pronin et al., 2004), we asked the respondents to share their perceptions of fake news and how these perceptions lead to actions, such as fact-checking or sharing. Our coding process (described below) was used to glean insights into the potential biases that influence the respondents' perspectives on fake news.

The study was conducted in three stages: 1) survey design, 2) participant selection, and 3) coding and validation, with the analysis performed at each stage outlined in Figure 3.

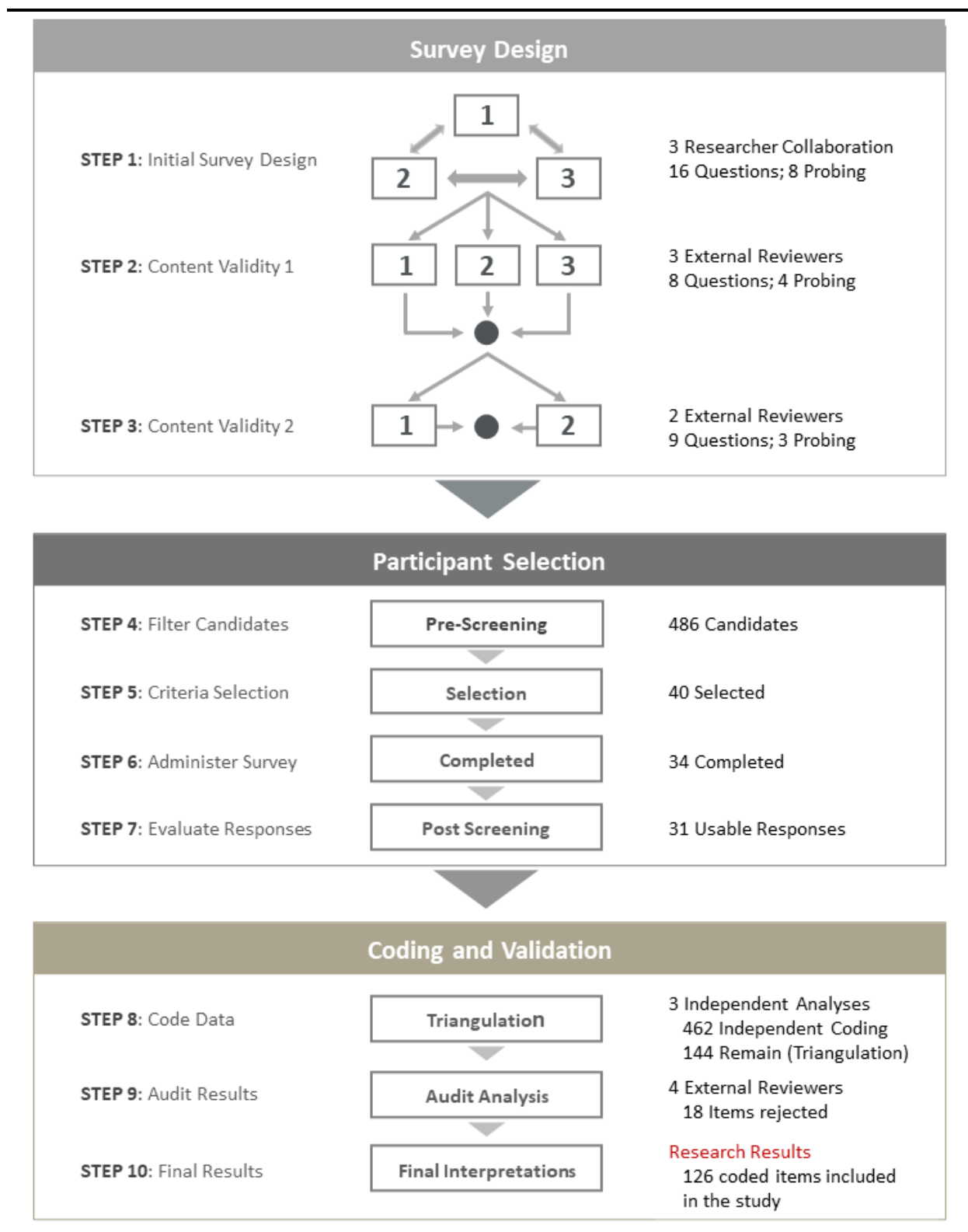


Figure 3. Research Design

3.1 Survey Design

Qualitative studies utilize non-probability samples to collect data from participants to evaluate a phenomenon, rather than generalizing to a population (Koerber and McMichael, 2008). The initial survey instrument was designed to capture user perceptions of truth with respect to sharing fake news on social media. It was divided into two parts with questions related to: 1) general sharing habits on social networking sites; and 2) gauging efforts made to verify news stories. Two rounds of content validity ensured the questions allowed the users to answer freely. The first round included three external reviewers and resulted in changes to the wording of several questions and recommendations for removing similar questions. A second round was conducted with two additional external reviewers. Another question was removed, one question edited to prevent leading the participants, and two questions added to capture perceptions about specific cases related to fake news. After two rounds of content validity, the final survey (Appendix A) consisted of 9 questions and 3 probing questions.²

3.2 Participant Selection

A web-based survey guided participants through each question and prompted probing questions to elicit more information when necessary. Our objective was to obtain participation from 30 people, since 20-30 participants are generally recommended for qualitative interviews (Marshall et al., 2013). Data was collected using a purposive sampling technique, designed to identify participants that match the qualities desired to study a phenomenon (Koerber and McMichael, 2008). Facebook was used since it is the leading social media source for shared news (Watson, 2020). To identify candidates for participation, we started with a convenience sample of Facebook connections from three Facebook accounts. Connections selected as potential candidates included friends, family, co-workers, professional relationships, and students. The initial list of candidates included 486 subjects who were selected based on regular Facebook use (i.e., multiple posts per week).

² The intermediate versions are available from the authors.

We used a purposive sampling technique on the list of 486 candidates to identify users that matched the specific criteria needed to explore the phenomenon assessed in this study. That is, participants were included only if they regularly share, like, or comment on news stories, regardless of whether a story contains fake news. With a target of 30 participants for the study, we identified the 40 strongest candidates that matched the criteria for our study based on frequency of news sharing. Of the 40 candidates invited to participate, 34 responded. Three were removed for incomplete/missing responses, resulting in data from 31 participants in the final sample. Table 2 summarizes the demographic data.

Table 2. Demographic Data of Participants

<i>Age</i>	<i>Ethnicity</i>	<i>Education</i>
9 – 18-24 years old	16 – White	2 – High school or equivalent (e.g., GED)
6 – 24-34 years old	6 – Hispanic or Latino	8 – Some college credit, no degree
8 – 35-44 years old	2 – Black or African American	9 – Bachelor's degree
2 – 45-54 years old	4 – Asian / Pacific Islander	6 – Master's degree
2 – 55-64 years old	1 – Other	4 – Doctorate degree
2 – 65 years or older	2 – N/A	2 – N/A
2 – N/A		
<i>Gender</i>	<i>Location</i>	
14 – Male	1 – Northeast - CT, DC, DE, MA, MD, ME, NH, NJ, NY, PA, RI, VT	
15 – Female	14 – Southeast - AL, AR, FL, GA, KY, LA, MS, NC, SC, TN, VA, WV	
2 – N/A	12 – Southwest - AZ, NM, OK, TX	
	1 – West - AK, CA, CO, HI, ID, MT, NV, OR, UT, WA, WY	
	1 – Other Country (Japan (US citizen))	
	2 – N/A	

* n=31 total participants

The study sought to target 30 participants based on recommended guidelines for participation (Marshall et al., 2013). However, proper qualitative analysis depends on data saturation, which means that the addition of new participants reveals no new findings (Marshall et al., 2013). To assess data saturation, we revisited our analysis and coding after the survey was complete. The results of our coding (detailed below) yielded eight cognitive biases in the responses, five of which emerged as significant. Our post hoc analysis revealed that an occurrence of all eight cognitive biases were coded within the first six participants. Furthermore, of the three cognitive biases that did not emerge as significant, 9 out of 13 coded responses (70 percent) were obtained from the first 20 participants. Based on these results, it appears we reached data

saturation from our 31 participants with no new participants required to continue. There is, however, a potential for selection bias because of the purposive sample employed. Due to the exploratory nature of this study, the results cannot be generalized to the population of Facebook or other social media users, but rather serves as a foundation for future research.

3.3 Coding and Validation

To evaluate perceptions of fake news and identify the cognitive biases that influence believability during consumption and subsequent sharing, three independent raters coded the open-ended responses based on the ten cognitive biases in Table 1 (CFI, 2020). The purpose of the coding was to identify the presence of any of the 10 cognitive biases in the open-ended response from the participants. For example, one participant who reportedly does not fact check information online stated “I’d rather quickly look at a couple sites and if the consensus is the same, I generally believe it to be true.” This comment was coded as a herding bias due to the participant believing information based on a collective of information. The coding and validation processes were designed to ensure reliability based on consistency and to ensure accuracy of the results (Mingers and Standing, 2020). Triangulation and inter-rater reliability were implemented to achieve these goals.

Triangulation is defined as “the process of improving the probability of the findings through different means.” (Baskerville et al., 2015). Triangulation is needed because a single method provides only one perspective that could potentially include bias (Patton, 1999). Four types of triangulation have been identified in the literature as (Parikh, 2002):

1. Data source triangulation: Collect multiple instances of data from a single source or combine data from multiple sources of the same information.
2. Investigator triangulation: Comprise a team with multiple investigators to examine the same phenomenon to increase reliability of the results.
3. Theory triangulation: Apply multiple theoretical perspectives to study a phenomenon.
4. Methodological triangulation: Confirm propositions through two or more measures to increase confidence in interpretations.

Investigator triangulation was used to validate the coding of the cognitive biases (Baskerville et al., 2015), with inter-rater reliability used to ensure reliability when interpreting text (Boudreau et al., 2001). Three raters considered the list of ten biases and agreed on the operational definitions used for coding to ensure that they had a consistent understanding of each bias and the rating process (Mingers and Standing, 2020). Each rater then evaluated the participant responses and identified passages that could be linked to one of the ten biases on the list. All passages were captured, and the coded biases were saved in excel spreadsheets. The coded passages from all three investigators were then combined into a single file to conduct triangulation and compare the results. In some instances, one coder selected a portion of a sentence to be coded while another highlighted the entire sentence. Although the length of the passage recorded by raters sometimes varied, the passages that were coded are the same so the cognitive biases recorded could be evaluated.

The evaluation process of coding by the three raters required matching captured phrases and the associated cognitive biases. In instances where all three raters identified the same phrase and coded the same bias, the results were recorded. Instances where only one rater identified a phrase or cognitive bias were discarded. Any phrases or cognitive biases identified by two coders were then discussed in depth for consideration to be included in the results in case the third coder missed the coding or did not agree. If consensus was reached, the coded phrase was included in the results. Otherwise, it was discarded.

While validity of the coding can be established through triangulation, it can be further strengthened through external audits (Cypress, 2017). This research implemented an additional layer of validity using external auditors who were not members of the original coding team. The external auditors were familiar with fake news research and cognitive biases. They were trained in the coding procedures and the operational definitions of the cognitive biases were discussed to create a consistent understanding among the auditors. Four external auditors independently evaluated the results, rating each coded item on a 5-point Likert scale, ranging from strongly disagree to strongly agree. The results were triangulated and instances where at least three out of four auditors rated a coded cognitive bias as either disagree or strongly disagree were removed.

4. Analysis and Results

The initial rating by the three investigators yielded 462 total coded items that were combined and triangulated. Each item consisted of a coded passage that was matched among the investigators' coding. For example, two investigators identified the following passage as containing a cognitive bias: "I'd rather quickly look at a couple [of] sites and if the consensus is the same, I generally believe it to be true.". However, the third investigator only included a subset of this passage: "if the consensus is the same, I generally believe it to be true." All three investigators coded the passages as *herding bias*, so the full passage was included in the final results. Passages were matched and combined with the most complete version of the passage. The coded items were assessed by the primary investigators for inter-rater reliability. Any coded passages that were only identified by a single investigator were discarded from the results. Passages with consistent coding by all three investigators were retained. The three primary investigators then discussed any passage where only 2 out of 3 investigators coded a cognitive bias. If consensus was met, the passage and associated bias was included; otherwise, the coded item was discarded. This process yielded 144 coded responses.

Next, external auditors underwent a training exercise to ensure proper understanding of the cognitive biases and task to be completed by rating 10 items. For example, the auditors each rated their level of agreement with the investigators' consensus that the following passage should be coded as *herding bias*: "I'd rather quickly look at a couple [of] sites and if the consensus is the same, I generally believe it to be true." For this passage, all four auditors either "agreed" or "strongly agreed," so the passage and associated coding was retained. After successful training, each external auditor independently evaluated all passages coded by the investigators, rating each coded item on a 5-point Likert scale, ranging from strongly disagree to strongly agree. The results were triangulated and all coding where three or four auditors disagreed with the coding were removed.

The final coding consisted of 126 items assigned to 8 of the cognitive biases. Table 3 shows the results, with the coding counts summarized and the cognitive biases with the highest counts identified. From this

list, five cognitive biases emerged as the most common. A list of all coding for the cognitive biases is provided in Appendix B.

Table 3. Qualitative Coding Results

Initial Coding:	462	Auditor Reduction:	18
Triangulated Coding:	144	Final Coded Count:	126

Rater Triangulation Results		Auditor Triangulation Results	
HM	27	HM	25
CB	27	CB	23
FCB	25	FCB	23
OB	24	OB	22
AB	22	AB	20
NF	13	NF	8
RH	3	RH	3
LA	3	LA	2
HS	0	HS	0
SS	0	SS	0

<i>AB: Anchoring Bias</i>	<i>LA: Loss Aversion</i>
<i>CB: Confirmation Bias</i>	<i>NF: Narrative Fallacy</i>
<i>FCB: Framing Cognitive Bias</i>	<i>OB: Overconfidence Bias</i>
<i>HM: Herd Mentality</i>	<i>RH: Representativeness Heuristic</i>
<i>HS: Hindsight Bias</i>	<i>SS: Self-serving Bias</i>

In addition to coding cognitive biases, we evaluated perceptions of fake news and user habits based on the two sections of the survey (general sharing habits and fake news perceptions).

4.1 General Sharing Habits

Our survey first elicited general news sharing habits (questions 1-3) so we could compare the answers with questions 6-8, which specifically asked about sharing fake news. By asking questions in both contexts, we create a base-level of understanding of the participant's general news sharing habits that can be contrasted with their awareness of fake news sharing behaviors.

The participants reported sharing news stories on social media an average of 1 -3 times per week. Social networking newsfeeds were the most prominent source of news consumption and sharing (24 participants). Only 3 participants reported they shared fake news intentionally whereas 27 stated they had never done so. The 3 participants who indicated they had shared fake news intentionally cited "entertainment," "to get a laugh," and "satire" as reasons. Of the 31 participants, 21 reported that they did not believe they had ever

unintentionally shared fake news. Six participants acknowledged that they had unintentionally shared fake news they perceived to be true at the time, but took action later to alert others or remove the shared news once it was deemed fake. The results from both sections of the survey demonstrate that people generally believe they, personally, are not contributors to fake news spread.

The third question asked participants about their purpose for sharing a news story. Two major themes emerged. The first revolved around participants reporting that they shared information because they found it to be personally interesting or funny, or because they liked it. One participant stated, *“If I see something funny I will share,”* whereas another said they share if *“news is interesting to me.”* The second major theme was that the participants believed the content (or story) had importance for others. One participant stated they share in order *“to spread awareness to those around me,”* whereas another stated they share information *“that I think will be informative to those who follow me.”* These types of responses demonstrated strong levels of confirmation bias and overconfidence bias because participants chose to share information that either confirmed their ideas, or reflected a belief they thought they were an authority on, or that they believed to be factual knowledge. Anchoring bias was prevalent among the responses related to reasons for sharing because participants indicated that the information shared was self-confirmed based on pre-existing knowledge, or confirmation was sought through specific, trusted sources. A variety of other reasons for sharing third-party source materials were identified by a smaller number of participants, including: stimulating conversation, educating others, expressing outrage, and sharing unusual or lesser known news.

4.2 Fake News Perceptions

The second section of the survey (questions 4, 5, and 8) asked questions directly related to fake news with respect to sharing habits, fact checking, and outcomes of fake news. We asked participants if they fact check before sharing on social media. Of the 31 participants, 10 reported fact checking before sharing; 21 participants reported they did not. When the 10 were asked why they fact checked, there were three common answers: 1) they did not want to contribute to the amount of fake news that already exists; 2) they wanted to ensure the information was accurate and trustworthy; and 3) they were concerned about their reputation.

The 21 participants who indicated they do not fact check were asked why not. Common themes were: 1) they believed they had the ability to distinguish fake from real news; 2) they were sharing news from sources they deem credible; and 3) they were not concerned or willing to put in the effort. The cognitive biases attributed to these actions by the coders were overconfidence and anchoring.

When asked how participants assessed whether third-party source information was real or fake, common replies included comparing news against sources the participants deemed credible, validating news through online searches, or self-identification and the ability to assess when news is fake. Anchoring bias was prevalent with one participant stating they “*Search news using similar terms used in the news,*” whereas another responded, “*If I see it on different sites/platforms I tend to believe it.*” Overconfidence bias was demonstrated by a participant who stated they can identify if news is real “*If it looks credible;*” another stated, “*I view different news sources, and draw my own conclusions.*”

The participants were then asked about their perceptions of fake news, in general, and their opinion on societal outcomes due to the spread of fake news. Several argued that fake news is used for power, attention, or monetary benefits and can cause people to be ill-informed. One stated, “*people who create fake news do it for attention*” whereas another stated: “*Fake News is used intentionally to misinform people.*” Regarding the societal effects of fake news, responses revolved around creating distrust among the media, creating a false narrative causing people to believe inaccurate information, and creating fear or panic. Comments related to the question involving societal effects appeared to be attributed to herd mentality and framing cognitive bias. One participant stated that fake news “*is shared to the point you start believing [it]*” and another stated that “*the info was wrong and could have hurt someone.*”

4.3 Cognitive Biases

As Table 3 shows, five cognitive biases had a higher frequency than others. Comments that displayed herd mentality bias were made the most often by the participants who suggested that fake news is shared because users follow the crowd or consensus. One participant stated: “*I'd rather quickly look at a couple [of] sites and if the consensus is the same, I generally believe it to be true.*” In other words, if fake news is shared by enough people, others start to perceive it to be true. This implies a network effect resulting in a herding bias

during the consumption of fake news where, the more prevalent a fake news story, the more people acknowledge and continue to share, like, or comment on it. Framing cognitive bias (FCB) also emerged as an important influencing factor of perceived truth and the spread of fake news. One participant stated *“sensationalized reporting with COVID-19 on one end has created panic buying,”* demonstrating that the way in which news stories were framed resulted in extreme actions. Participant responses also indicated that framing cognitive bias may have contributed to continued infections of COVID-19 and death because people believed it was not something to take seriously, or that alternative treatments would help fight the virus.

Overconfidence was also relevant. Many participants believed they had the ability to identify fake news and blamed others for the fake news problem. Sample responses to questions asking how they could identify whether third-party news is real or fake were: *“If it looks credible,” “draw my own conclusions,”* and *“in most cases people should be able to make the distinction between real and fake by simply pausing and using their brain.”* These participants were confident in their ability to identify fake news by simply reading it.

Consistent with prior research, confirmation bias emerged as impactful for the believability of fake news. This was evident when participants were not concerned with identifying whether news is real or fake, but rather believed and shared stories that agreed with their point of view. Examples include: *“Share information that is important to me”* and *“I often only share things that highlight his utter ineptitude, racist views, or psychopathic behavior.”* From this cognitive bias, people seek out information that confirms their belief regardless of accuracy.

The fifth common cognitive bias that emerged was anchoring. Responses demonstrating this behavior include *“I don't think it's necessary to fact check articles from CNN”* and *“fake news can be avoided by only viewing news sources I have used for years and trust.”* In these examples, the participants anchored their beliefs to the source of the information, rather than the truthfulness of the content. While anchoring bias in this example stems from the view of “expert-opinion” anchoring, consumers of information may also anchor to specific pieces of information within a news article they believe to be true causing them not

to question whether other information in the article is true. Therefore, the source, the expert-opinion of the author, or a specific piece of information may be associated with anchoring cognitive bias.

4.4 Summary

Our analysis aggregates descriptive results and interprets qualitative results as summarized in Table 4.

Table 4. Summary of Findings

Frequency of sharing news online: 1-3 times per week			
News sources used for sharing content			
Social networking news feed	24	Online news publications	14
Internet search	20	Other	5
Shared fake news intentionally		Shared fake news unintentionally	
No: 27	Yes: 3	No: 21	Yes: 6
10 participants reported that they fact check information before sharing			
did not want to contribute to the amount of fake news that already exists			
wanted to ensure the information was accurate and trustworthy			
were concerned about their reputation			
21 participants reported that they do not fact check information before sharing			
ability to distinguish fake news from real news			
sharing of news from sources they deem credible			
not concerned or willing to put in the effort			
Cognitive biases influencing fake news consumption			
Herd Mentality			
Framing Cognitive Bias			
Overconfidence Bias			
Confirmation Bias			
Anchoring Bias			

**Based on 31 participants*

This analysis showed that most participants did not intentionally share fake news; however, they did not take active steps to ensure that all content they share was truthful. Biases influenced the conviction level pertaining to whether users could identify fake news and/or avoid sharing it. Importantly, five cognitive biases emerged as significant influencers of believability, and subsequent sharing: herd, framing, overconfidence, confirmation, and anchoring.

5. Discussion

This research conducted an exploratory analysis to identify cognitive biases that may influence the believability of fake news. Such biases often result from data-related manipulations, obscurities, or a lack

of effort to evaluate or verify information (Emby and Finley, 1997). In our study, two-thirds of the participants reported that they do not fact check information before sharing, presumably because they believe the information (Schuetz et al., 2021). Cognitive biases also influence one's deception detection capabilities (Millar and Millar, 1997), making someone more susceptible to believing fake news.

Together, cognitive biases and the lack of effort by a consumer to verify the truthfulness of a story create an environment that increases the believability of fake news. When fake news consumers neglect to verify information (Torres et al., 2018), they must rely on personal knowledge and their own assessment of whether a piece of news is internally consistent and logical (coherence). A lack of effort can also result in reduced attention as consumers scower articles, allowing cognitive biases to influence believability, and subsequent actions (Cheng and Wu, 2010). Cognitive biases result in poor decision making because individuals are typically unaware of their own biases (Lee et al., 2016) and unable to logically evaluate truth as coherence theory suggests (Dunwoody, 2009). Although it is not possible to control human efforts when consuming content, cognitive bias mitigation can assist consumers in using both correspondence and coherence when evaluating believability.

Our analysis revealed perceptions of fake news, behaviors, and cognitive biases that influence believability during news consumption. Most participants reported that they often consume content through social media, with two-thirds acknowledging no effort to verify or validate the content they consume and share. Without correspondence to evaluate truthfulness, consumers might rely only on the coherence of a story, thus allowing fake news creators to target the cognitive biases that influence believability. Therefore, proactive measures should be identified to minimize the effects of cognitive biases on the believability of fake news and reduce the adverse effects that occur through consumption and subsequent engagement behavior (i.e., like, share, comment), which contribute to its spread. For each cognitive bias identified in our analysis, we recommend mitigation methods, grounded in literature. These recommendations are based on existing studies and proven techniques to minimize the effects of each cognitive bias on believability. We apply these mitigation methods to the context of fake news to explain how they could potentially help reduce the believability of fake news. Not all cognitive biases are mutually exclusive, so some mitigation

methods could address multiple biases. Based on the mitigation methods identified, we make recommendations for future research to empirically evaluate each method to validate its effectiveness.

The Cognitive Bias Mitigation Model in Figure 4 identifies seven mitigation methods we propose to address the biases. They map into three overall categories: mitigation groups, mitigation methods, and cognitive biases.

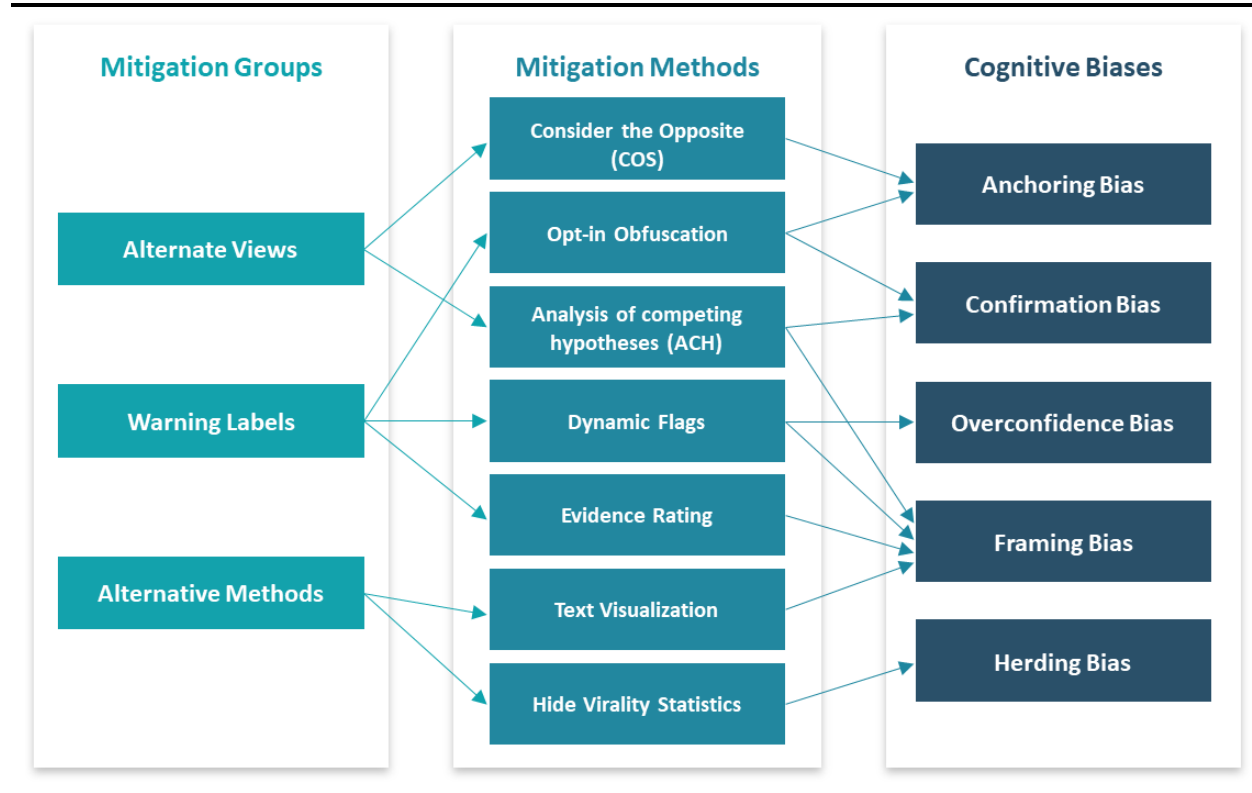


Figure 4. Cognitive Bias Mitigation Model

5.1 Mitigation Group: Alternative Views

In an information scarce environment, individuals tend to anchor their beliefs to the information available (Adame, 2016). In environments with an abundance of information (e.g., the Internet), users may be overwhelmed by the amount of information available, thereby reducing their ability to identify truth (Lorenz-Spreen et al., 2020). The abundance of information choices results in a '*paradox of information access*' where individuals consume more content from similar ideologies (Abdelzaher et al., 2020). Biases may be evident when individuals anchor their beliefs to the news they consume or tend to believe news that

confirms their beliefs. *Alternative views*, such as a consider-the-opposite strategy or an analysis of competing hypotheses, can counteract cognitive biases by presenting other views or hypotheses.

5.1.1 Consider-the-Opposite Strategy

A consider-the-opposite (COS) strategy (Adame, 2016) has been successfully applied to information scarce environments where users tend to anchor their beliefs to the information available. On social media sites, users are often presented with information that aligns with their ideology. By providing opposing information, anchoring bias may be effectively mitigated, allowing users to process both the information they have along with opposing information. This method applies correspondence and coherence criteria for truth-seeking behaviors.

Recommendation #1:

News articles appearing in a feed could also link to articles with opposing or opposite views, providing a concrete, alternative perspective.

5.1.2 Analysis of Competing Hypotheses

Making people aware of their confirmation bias does not mitigate their bias (Lehner et al., 2008). The Analysis of Competing Hypotheses (ACH) is a confirmation bias mitigation technique where competing hypotheses are presented requiring the consumer to disconfirm evidence to create a coherent story based on corresponding evidence (Lehner et al, 2008). The ACH method has been effective in reducing confirmation bias in individuals who are not highly experienced in an area (Lehner et al., 2008), and may also have a compounding effect when used in conjunction with COS. Combining these methods, individuals can be presented with opposing information and competing hypotheses to stimulate critical thinking. In this way, ACH could reduce framing bias because individuals focus on corresponding evidence, rather than on how information is framed.

Recommendation #2:

ACH can present competing hypotheses along with the articles presented to stimulate thinking and engage the consumer to disconfirm information as they consume it.

5.2 Mitigation Group: Warning Labels

Warning labels in the form of fake news flags have limited effects on influencing believability and cognitive biases, especially confirmation bias (Moravec et al., 2022, Moravec et al., 2019). The use of mitigation measures such as flags, source ratings, and fact checking have yielded inconsistent results (George et al., 2021), although warning labels might be useful for targeting certain cognitive biases. A wide variety of types of warning labels, such as obfuscation, dynamic flags, and evidence rating, could play a significant role in mitigating the cognitive bias effects of anchoring, confirmation bias, overconfidence bias, and framing bias (Rieger et al., 2021, Schall et al., 2017).

5.2.1 Opt-in Obfuscation

Obfuscation makes something obscure or unclear. In contrast to general warning labels or flags, obfuscation provides a warning label that the user must click prior to viewing the news story. Opt-in obfuscation has been applied to information-seeking efforts warning users of potential confirmation bias, obscuring information until the user clicks a button to accept the risk (Rieger et al., 2021). Rieger et al. (2021) found that obfuscation reduced confirmation bias by encouraging consumers to view both attitude-confirming and attitude-opposing information. They recommend using an opt-in approach to increase the consumption of attitude-opposing information. Obfuscation might work in conjunction with COS to further reduce anchoring bias and in conjunction with ACH to strengthen the effect of confirmation bias mitigation.

Recommendation #3:

Obscure attitude-confirming information and require users to click a button to knowingly accept risk.

5.2.2 Dynamic Flags

Although flags and warnings can assist in identifying fake news, the results of their effectiveness are mixed (George et al., 2021). From our analysis, it is evident that overconfidence bias is problematic for consumers of fake news.

Static warning flags (e.g., signs, labels, and text) have been inconsistent in mitigating overconfidence bias, thus having limited results in identifying and reducing the spread of fake news (George et al., 2021). However, dynamic flags consisting of changing stimuli or the required interaction have demonstrated a

positive impact in mitigating overconfidence bias (Schall et al., 2017). A dynamic warning could consist of attention-grabbing content, such as blinking warning labels or completely covering the content and requiring user interaction (e.g., clicking a button) to view an article (Schall et al., 2017). Facebook implemented dynamic warning labels alerting users that: content has been identified as fake news; or a page has repeatedly shared false information (Mukul, 2021). This requires the user to click a button to remove the label and read the article (Mukul, 2021). Further research is needed to evaluate the effectiveness of reducing overconfident bias using dynamic warnings. Furthermore, dynamic flags could work in conjunction with other mitigation methods (e.g., ACH, text visualization, evidence rating) because they can increase attention to the mitigation methods being employed to reduce framing bias.

Recommendation #4:

Incorporate the use of dynamic flags in addition to, or in place of, static flags.

5.2.3 Evidence Rating

Evidence rating requires individuals to evaluate and rate the information they consume. Moravec et al (2022) evaluated the use of credibility ratings to combat cognitive bias when consuming social media news. Data-related issues or a lack of effort on the consumer's part can lead to framing bias (Emby and Finley, 1997). Moravec et al. (2022) evaluated the use of credibility ratings, but with respect to confirmation, rather than framing, bias. Although credibility ratings did result in consumers critically thinking about the truthfulness of articles, they were ineffective for mitigating confirmation bias (Moravec et al., 2022).

Evidence rating has long been used to evaluate its effects on framing bias (Emby and Finley, 1997). One evaluation demonstrated that evidence rating was able to remove the framing effect as users focused on the data, rather than how it was framed (Emby and Finley, 1997). In 2018, Facebook launched a credibility rating system (Dwoskin and Shaban, 2018). However, the functionality was misused because users reported content with which they did not agree (i.e., biased against), causing Facebook to launch additional functionality for rating the trustworthiness of the raters (Stewart, 2018).

There are many examples where the term 'fake news' has been employed to attack opposing opinions (Tong et al., 2020). The evidence rating functionality should be applied to rate content, rather than

report content for a self-serving objective of removing it. Removing the ability to weaponize evidence rating functionality would decouple it from confirmation biases, allowing it to mitigate framing bias.

Recommendation #5:

Apply evidence rating functionality to rate content, not remove it.

5.3 Mitigation Group: Alternative Methods

Two alternative methods are text visualizations to mitigate framing bias and hiding virality statistics to mitigate herding bias.

5.3.1 Text Visualization

Stimulating critical thinking through ACH (Lehner et al., 2008), using dynamic flags (Cheng and Wu, 2010), and rating evidence have been used to mitigate framing bias by alerting users of false information or stimulating critical thinking regarding the content being consumed. Text visualization seeks to remove the focus on specific words prior to allow the consumer to consider all information being presented, rather than focusing on how it is framed (Baumer et al., 2018).

Word clouds are text visualizations depicting word frequency with high importance of words displayed in larger text. An example is presented in Figure 5. Word clouds influence cognitive and interpretive processes during consumption of information (Baumer et al., 2018). This visualization focuses on the disruption of a linear structure of text used to create framing effects. It can be applied to social media using natural language processing and AI algorithms to generate word clouds based on articles that are presented along with their headlines. A visual display of word could alert users of information contained in an article to reduce framing effects.



Figure 5. Word Cloud of Discussion Section

Recommendation #6:

Employ text visualization to alert consumers of information contained in the article.

5.3.2 Hide Virality Statistics

Virality metrics on social media influence correspondence truth as increased numbers of likes, comments, and shares positively influence believability of information (Kim, 2018). This supports a herding bias as individuals believe content that has a high level of engagement, based on virality metrics. Correspondence truth poses a significant challenge in addressing herding bias because it would require addressing group level behavior or mitigating herding effects.

Recommendation #7:

Social networks should hide virality metrics for news articles to mitigate herding bias.

5.4 Contributions and Limitations

This research provides several theoretical, methodological, and practical contributions.

5.4.1 Theoretical Contributions

This research identified cognitive biases that influence the believability of fake news leading to its spread; and proposed a model of mitigation measures for each bias. The research demonstrates the complexity of cognitive bias mitigation and the potential for multiple cognitive biases to be addressed simultaneously.

We propose a Cognitive Bias Mitigation Model comprised of mitigation methods to influence five cognitive biases and reduce believability of fake news.

5.4.2 Methodological Contributions

This research contributes to efforts to increase rigor and reliability in information systems research by introducing recommendations for identifying and mitigating fake news through assessing the biases inherent in its believability. From an interpretive view, dependability improves reliability and replicability, with rigor focusing on internal consistency (Gasson, 2004).

For qualitative research, inter-rater reliability (IRR) evaluates the level of agreement to ensure consistency, enable replicability, and reduce researchers' confirmation bias (McDonald et al., 2019). The latter is relevant for this research because it extends coding rigor by introducing an additional form of reliability referred to as *code auditing*. We conducted IRR to identify cognitive biases by rating coded quotes. Upon completion of our analysis, we used four external researchers to evaluate and rate each quote on a Likert scale. The external reviewers were briefed on the definitions of the cognitive biases coded in the initial analysis and their ratings evaluated to determine internal consistency among the external auditors. This resulted in removing several coded quotes and increasing the reliability of our results.

5.4.3 Practical Implications

This research has significant practical implications for combating fake news. We provide a psychological understanding of the cognitive biases that influence believability. This can help social media networks to identify and understand factors that influence believability to better mitigate fake news and minimize its resulting impact. The recommended mitigation methods could be employed to assist users in identifying fake news and reducing its believability.

5.4.4 Limitations and Future Research

This research has some limitations that we recognize and opportunities for future research. First, this research is exploratory in nature, so the results may have limited generalizability to all social media users and fake news contexts. However, the insights provided demonstrate additional cognitive biases that can influence believability of fake news that future research can build on. For example, future research should

empirically validate the cognitive biases influence on the relationship of consumption and believability and generalize the findings to larger populations and various social media platforms.

Second, we identified five cognitive biases that were influential among our study participants in a fake news context, but the list of ten cognitive biases (Table 1) evaluated are not exhaustive. Because of the lack of a comprehensive list of cognitive biases grounded in the literature, we employed a list of cognitive biases from a practitioner source, and augmented with academic sources, that focused on believability of deceptive information. It is possible that additional cognitive biases may be influential within the context of fake news believability and should be explored in future research.

Third, the survey used self-reported data, although we attempted to minimize concerns related to faulty self-perceptions by asking respondents to describe their perceptions and thoughts and coding that data to identify evidence of bias. Future studies that can more objectively obtain data would be useful. More specifically, experimental research could empirically be conducted in a controlled environment to evaluate each cognitive bias and its influence on believability empirically.

Lastly, the mitigation recommendations are based on findings from prior research, demonstrating successful measures in reducing believability. These measures are applied to the cognitive biases identified in this research, but not tested for their effectiveness. Future research should conduct experiments and empirical research to evaluate the effectiveness of the recommended mitigation methods in reducing believability of fake news.

6. Conclusion

This research has investigated the influence of biases on the believability of fake news. We conducted a qualitative survey on social media users' perceptions of fake news from which five cognitive biases emerged as being influential in moving social media users from coherence to correspondence truth. These are: herd, framing, overconfidence, confirmation, and anchoring. We propose a Cognitive Bias Mitigation Model comprised of three mitigation groups (alternative views, warning labels, and alternative methods) to identify seven mitigation strategies. Further research is needed to examine additional cognitive biases and mitigation methods.

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Appendix A. Survey Questions

The operational definitions of third-party news and fake news were provided to participants as they answered the following questions.

General Sharing Habits

Third-Party Source: A source providing content (i.e. information, news, or stories) that is not created by the social networking user or social networking platform.

1. How often do you share content on a social networking platform that comes from a third-party source?
 - a. Less than once per week
 - b. 1-3 times per week
 - c. 4-6 times per week
 - d. Once per day
 - e. Multiple times per day
2. Which third-party sources have you used for sharing content? (check all that apply)
 - a. ☐ Social networking news feed
 - b. ☐ Internet search
 - c. ☐ Online news publications
 - d. ☐ Other (please list _____)
3. What is your typical purpose for sharing third-party source content?

Fake News Related Questions

Fake News: Articles that look like news stories but contain misleading, false, inaccurate, or biased information. These types of articles are often shared through social networking platforms.

4. Do you fact check third-party source materials before sharing?
 - a. If Yes, why?
 - b. If No, why not?
5. How do you determine whether third-party source information is fake or real news?
6. To the best of your knowledge, have you ever shared fake news intentionally? (Y/N)
 - a. If Yes, why?
7. To the best of your knowledge, have you ever shared fake news unintentionally? (Y/N)
8. What action would you take if you discovered that third-party source material you shared unintentionally contained fake news?
9. In your opinion, what are the societal effects of fake news?
 - a. Can you provide examples?

Demographic Data

Age

- 18-24 years old
- 24-34 years old
- 35-44 years old
- 45-54 years old
- 55-64 years old
- 65 years or older

Ethnicity

- White
- Hispanic or Latino
- Black or African American
- Native American or American Indian
- Asian / Pacific Islander
- Other

Education

- Some high school, no diploma
- High school or equivalent (e.g., GED)
- Some college credit, no degree
- Trade/technical/vocational training
- Associate degree
- Bachelor's degree
- Master's degree
- Doctorate degree

Gender

- Male
- Female
- Transgender Male
- Transgender Female
- Non-conforming
- Other
- Prefer not to answer

Location

- Midwest - IA, IL, IN, KS, MI, MN, MO, ND, NE, OH, SD, WI
- Northeast - CT, DC, DE, MA, MD, ME, NH, NJ, NY, PA, RI, VT
- Southeast - AL, AR, FL, GA, KY, LA, MS, NC, SC, TN, VA, WV
- Southwest - AZ, NM, OK, TX
- West - AK, CA, CO, HI, ID, MT, NV, OR, UT, WA, WY
- Other Country (Please Specify)

Appendix B. Cognitive Bias Coding Results

Each coded quote may appear under one or more cognitive biases.

Question	Response
<i>Herd Mentality (25)</i>	
Why do you not fact check third-party source materials before sharing?	I'd rather quickly look at a couple sites and if the consensus is the same, I generally believe it to be true.
How do you determine whether third-party source information is fake or real news?	If I see it on different sites/platforms I tend to believe it.
In your opinion, what are the societal effects of fake news?	in some cases, it causes unnecessary panic, leads to poor decision-making in businesses/organizations/communities, and several other related snowball effects is shared to the point you start believing Essentially it promotes an "anti-establishment" bias where doctors and scientists are not trusted leading to poor decisions for society. Depending on the topic, it can quickly create panic or fear, or even be a danger to some people. There is a general culture of mistrust when it comes to media especially with political biases. some of the crazy issues with coronavirus are sparked by fake news. It is incredibly impactful, especially because of the many platforms and high engagement rate of modern generations. It seems to spread quicker than real news, sometimes. Fake news is entertaining the masses at times.
Can you provide examples of the societal effects of fake news?	In Russia they have bots that all day post on social media, and once was able to organize two protest on the same day in America For example false or sensationalized reporting with COVID-19 on one end has created panic buying and xenophobic attacks, while on the other end has people taking no precautions at all. People I know who are republicans will claim fake news when you don't agree with their opinion. People I know who are democrats will claim racism if you don't agree with their opinion.
What is your perception of fake news?	fake news will always be effective because of people's unwillingness to educate themselves on topics I think the problem is social media because people continually share this fake stories and it spreads so fast. People tend to believe everything they read on the internet. Even if you don't share it, you can still see if someone has liked or commented on an article making it spread to others even more.
In your opinion, what effect does fake news have on elections (president, congress, senate, etc)?	it confuses people and could sway the masses towards or away from a political faction It has a big effect on all elections, it could harm an election. The smaller the scope of the election (federal->state->county->city->district), the larger the effect. Seems like herd-mentality to get willing folks to vote as some want. Fake news if left unchecked can sway large groups of people and change an election Can completely sway an election
In your opinion, what effect does fake news have on the current Coronavirus (COVID-19) pandemic?	This has caused for huge fear in people causing them to buy in extreme amounts. There is a lot of fake news that scares people and there are so many it is hard to tell what is real or fake, it causes panic. I think fake news has drowned out the information provided by actual experts - and consequently, many are acting on poor or unproven advice. Many of my friends shared these stories and said the media is just trying to scare people and Coronavirus is not serious. Because of this it has spread all over the world and now I have to stay at home in quarantine for weeks and who knows how much longer. For other people, they read stories about grocery stores running out of toilet paper, hand sanitizer and food. There are so many fake news spreading in the people regarding the fake news. People were panicking because of the news because of which people were stocking the toilet papers and other items, so there was a shortage.

Confirmation Bias (23)

What is your typical purpose for sharing third-party source content?

For sharing my opinions with others

to advise my friends of something important

Share information that is important to me and that I think will be informative to those who follow me.

I think it's either an interesting or helpful information/content related to everybody's concern.

To educate people on a topic I feel that they are misinformed on.

To share information that in my opinion is worth sharing, either to educate or inform.

To uplift or educate or remember ...

To inform others.

I often only share things that highlight his utter ineptitude, racist views, or psychopathic behavior.

To inform others

I also share information that I think it important.

Inform a friend of a headline or update that is relevant to us.

Why do you fact check third-party source materials before sharing?

I want to make sure that I am not sharing something that people can easily argue against, I want to be right, and I need the information to be true in order for me to be right.

In your opinion, what are the societal effects of fake news?

People are misinformed and people will not research information leading to them being misinformed but I think people don't care.

People tend to consume the news that agrees with their ideology, so fake news serves as a crutch people can use when they are wrong.

people LOVE confirmation bias. They blindly share whatever whatever fits their view of the world.

Can you provide examples of the societal effects of fake news?

People across the political spectrum tend to consume news from their favorite source, which is usually ideologically biased.

People I know who are republicans will claim fake news when you don't agree with their opinion. People I know who are democrats will claim racism if you don't agree with their opinion.

What is your perception of fake news?

People are seeking validation everywhere regarding a large variety of topics daily

People will continue to do it, however, because they LOVE to feel 'right' or like they are 'winning'.

In your opinion, what effect does fake news have on elections (president, congress, senate, etc)?

People already have established who they are gonna vote for and no amount of fake news will change their opinion

People felt she committed a crime and fake news supported people's beliefs. Even if people knew it was fake they didn't care, they shared anyway because they wanted to share anything that agreed with their opinion.

In your opinion, what effect does fake news have on the current Coronavirus (COVID-19) pandemic?

People will die due to fake news. States (China, NYC, etc.) that are hot spots have a vested interest in collecting cures so they will produce public perceptions that those cures don't work.

Framing Cognitive Bias (23)

In your opinion, what are the societal effects of fake news?

It has a big impact on our society, people can believe some crazy ideas and it can be harmful to others and the environment.

Causes panic and fear. Changes normal behavior

Leads to panic and gives people false hopes

it can quickly create panic or fear, or even be a danger to some people. For example, the fake news of vaccines being dangerous has been something that has harmed communities and disparate populations.

some of the crazy issues with coronavirus are sparked by fake news. It is incredibly impactful, especially because of the many platforms and high engagement rate of modern generations.

Can you provide examples of the societal effects of fake news?

covid-19 pandemic; the fake news that it is harmless to younger generations and that they can carry on with their lives as if nothing was wrong
For example false or sensationalized reporting with COVID-19 on one end has created panic buying and xenophobic attacks, while on the other end has people taking no precautions at all.
What is your perception of fake news?
It's used to keep the masses unbalanced, so the powerful remain powerful. It allows the powerful to divide the consumers into opposing factions.
In your opinion, what effect does fake news have on elections (president, congress, senate, etc)?
I think fake news has little to do with elections because in my opinion the candidates are already feeding lies just to get support anyways. It's all about who can talk more and talk up to as many people as possible.
I think fake news has a very real effect on elections, and likely sways voters one way or another in an election.
It can skew election results and motivate voters to lean towards certain affiliations without logical reason.
I believe it can negatively or positively affect elections because of ignorance someone might see/read something and choose to vote or to not vote for that candidate based on fake news
In your opinion, what effect does fake news have on the current Coronavirus (COVID-19) pandemic?
This has caused for huge fear in people causing them to buy in extreme amounts.
There is a lot of fake news that scares people and there are so many it is hard to tell what is real or fake, it causes panic.
Panic buying; Unnecessary medical care
It's causing unfounded fear among some and a dangerous complacency among many others.
It can create more fear among people.
You have the president and some close to him making recommendations on specific medications people can take to lower their chances at contracting COVID-19. People have actually died as a direct result.
It has deep implications and I feel that a large part of the problem is that at a system level and leadership level of the nation, there was misinformation that was spread that many people took as fact.
Fake news on COVID19 can cause panic affecting mental health and recklessness increasing spread of the virus due to people not believing there is a problem worth caring about.
This caused panic and people started rushing to the grocery store and buying these things like crazy. This cause stores to really run out and made the panic grow even stronger.
People were panicking because of the news because of which people were stocking the toilet papers and other items, so there was a shortage.
It pushes people to the extremes of not doing enough or doing too much. Many people over panic and create chaos instead of control.

Overconfidence Bias (22)

What is your typical purpose for sharing third-party source content?
For sharing my opinions with others
Share information that is important to me and that I think will be informative to those who follow me.
I think it's either an interesting or helpful information/content related to everybody's concern.
To educate people on a topic I feel that they are misinformed on.
Share information to enlighten my friends or start a conversation.
To share information that in my opinion is worth sharing, either to educate or inform.
To inform others.
Informational, sharing research authorities I view as well-vetted
Generally share content to provide information to my peers as well as stimulate discourse among them.
Information for friends of interesting findings
To inform others
To spread awareness to those around me.
Why do you not fact check third-party source materials before sharing?
I fact check within the article. I can usually tell if it is contrived based on political ideology, social goals, or other.
If they look believable I share the information
I do only post from sites that don't appear to have an agenda

Why did you share fake news intentionally?
To misinform people and entertainment and to lead people to the correct information by debating the fake news.
How do you determine whether third-party source information is fake or real news?
If it looks credible.
Prior understanding of the subject matter combined with clues within the article.
I view different news sources, and draw my own conclusions
I only check if the website or source looks real.
In your opinion, what are the societal effects of fake news?
I find in most cases people should be able to make the distinction between real and fake by simply pausing and using their brain.
Can you provide examples of the societal effects of fake news?
Media attempting to report medically. I.e.. COVID 19 they have created widespread panic and anxiety over packaged goods, food, etc.

Anchoring Bias (20)

Why do you not fact check third-party source materials before sharing?
The best I do is only share from sources that I believe have little-to-no bias
I'd rather quickly look at a couple sites and if the consensus is the same, I generally believe it to be true.
I try to only share information from credible sources. I don't think it's necessary to fact check articles from CNN.
How do you determine whether third-party source information is fake or real news?
Search news using similar terms used in the news
Check sources and search for same news on other sites
news sources I have used for years and trust
I will click-on the content and read it first, and then go to online tools to search for published newspaper or news stories to confirm whether it's real or not.
Prior understanding of the subject matter combined with clues within the article.
Search google news to see if the story is shared by reliable news outlets and not just BS sites like natural news.
compare it with other articles that I have read from accredited news sources
If there are various sources stating what is said, it may be true.
First, I utilize google scholar. I try to find citations somewhere, and also I cross-reference the news with other reported material. I sometimes utilize fact-checking websites.
I determine if a third-party source information is real if there are several news articles with the same information.
I just google it and if the news is real, I am able to find some reliable source.
I google the same topic I read about and compare to other trusted primary sources.
Verify the same information from multiple and/or trusted sources
I also look to sites that I find are credible through historical uses/articles that held to be true.
Firstly by verifying the source of the news to be authentic. Then I will check the credibility/affiliations of the source to have awareness about political leanings.
I will also compare several different sources looking at each publisher's site
Can you provide examples of the societal effects of fake news?
People across the political spectrum tend to consume news from their favorite source, which is usually ideologically biased.

Narrative Fallacy (8)

In your opinion, what are the societal effects of fake news?
It negatively affects people's emotions and trust from one to another
I believe it can make a lot of society ignorant to what is really going on.
It seems to spread quicker than real news, sometimes. Fake news is entertaining the masses at times.
Can you provide examples of the societal effects of fake news?

As I mentioned before, there are several people who are taking advantage of people in vulnerable situations. For example, I saw a few news articles with people claiming to have a cure for COVID-19.

There's a statement that Obama didn't do anything for black people and doing some research there is an actual White House site dedicated to what Obama did for black people.

In your opinion, what effect does fake news have on elections (president, congress, senate, etc)?

cause people to believe things that are lies and produce certain feelings towards certain groups of people

Fake news makes it appear that politics are bad, or that republicans/democrats are doing something wrong when they aren't actually doing anything

Fake news allows candidates to appear better or worse than they are.

Representativeness Heuristic (3)

Can you provide examples of the societal effects of fake news?

someone was sharing fake news on COVID-19 and ways that people can fight and prevent it but the info was wrong and could have hurt someone

In your opinion, what effect does fake news have on the current Coronavirus (COVID-19) pandemic?

there are tons of miss information that is being shared that is based on speculation, like the heat kills covid19

I seen so many stories in January and February about how the flu has killed more people than Coronavirus.

Loss Aversion (2)

Why do you fact check third-party source materials before sharing?

It will affect my reputations

What action would you take if you discovered that third-party source material you shared unintentionally contained fake news?

I would never take the risk to share third-party news in the first place